

RawHash2 segmenter benchmark — Setup B (RawHash standard)

Eval = distance-based: TP iff same chrom and $|\text{center_query} - \text{center_truth}| \leq 10\text{kb}$.
This matches the RawHash/UNCALLED published evaluation (pafstats).

Why distance-based?

RawHash2 outputs the anchor-chain span (~ 500 bp), while minimap2 truth uses the full alignment span ($\sim 2\text{-}3$ kb). Even with perfectly correct mapping center, overlap-fraction of truth is only 5-20%. Setup A (overlap ≥ 0.1) penalizes this output-format gap; Setup B measures actual mapping correctness.

Metrics in this report

- $d \leq 10$ kb — primary (RawHash standard, see explanation page)
- $d \leq 100$ kb — lenient, useful for repetitive large genomes
- $\text{ovl} \geq 0.1$ — old metric (Setup A) — for cross-reference
- $\text{ovl} \geq 0.5$ — strict span-match (per-dataset table only) — output-format limit

Segmenters (9 — cusum dropped, algorithmically broken)

- default, scrappie (t-statistic family)
- pelt, binseg (DP change-point detection)
- hmm, bocd (probabilistic)
- window, mad (sliding-window)
- gradient (edge detection)

Datasets

D1 SARS-CoV-2 R9.4, D2 E. coli R9.4, D3 yeast R9.4, D4 green algae R9.4,
D6 E. coli R10.4, RB_ecoli R10.4.1, RB_dmel R10.4.1, RB_hsap_full GRCh38 R10.4.1.

Datasets (basic)

Dataset	Organism	Chemistry	Reads	Ref	FAST5
D1	SARS-CoV-2	R9.4	1,383	31 KB	11.00 GB
D2	E. coli	R9.4	89	5 MB	27.00 GB
D3	S. cerevisiae	R9.4	500	12 MB	0.40 GB
D4	C. reinhardtii	R9.4	500	112 MB	2.00 GB
D6	E. coli	R10.4	294	5 MB	98.00 GB
RB_ecoli	E. coli	R10.4.1	12,000	5 MB	0.93 GB
RB_dmel	D. melanogaster	R10.4.1	12,000	143 MB	0.73 GB
RB_hsap_full	H. sapiens (GRCh38)	R10.4.1	12,000	3.0 GB	7.10 GB

Datasets — extended characterization

DS	Organism	Chem	Reads	Ref bp	GC%	contigs	N50 (Mb)	N%	masked%
D1	SARS-CoV-2	R9.4	1,383	0.03 Mb	38.0	1	0.03	0.00	0.0
D2	E. coli	R9.4	89	5.23 Mb	50.5	1	5.23	0.00	0.0
D3	S. cerevisiae	R9.4	500	12.16 Mb	38.1	17	0.92	0.00	0.0
D4	C. reinhardtii	R9.4	500	111.10 Mb	64.1	53	7.78	3.65	32.4
D6	E. coli	R10.4	294	5.23 Mb	50.5	1	5.23	0.00	0.0
RB_ecoli	E. coli	R10.4.1	12,000	5.23 Mb	50.5	1	5.23	0.00	0.0
RB_dmel	D. melanogaster	R10.4.1	12,000	143.73 Mb	42.0	1,870	25.29	0.80	20.9
RB_hsap_fullH	sapiens (GRCh38)	R10.4.1	12,000	3.21 Gb	41.0	455	145.14	4.98	49.5

Ref bp = total reference length; *N50* = contig N50; *N%* = ambiguous N bases; *masked%* = soft-masked low-complexity / repeat content (lowercase in FASTA).

Primary metric explained — $|\text{center_q} - \text{center_t}| \leq 10 \text{ kb}$

What this means

Each PAF mapping (truth and query) defines a target-side interval $[\text{tstart}, \text{tend}]$ on the reference genome. The mapping CENTER is $(\text{tstart} + \text{tend}) / 2$.

A query mapping is True Positive iff:

- (1) same chromosome as truth's best mapping for that read, AND
- (2) $|\text{center_query} - \text{center_truth}| \leq 10\,000 \text{ bp}$.

If the read maps to a different chromosome, or the centers are more than 10 kb apart, it counts as False Positive (and the truth as False Negative). If truth exists but query has no mapping, it is False Negative. F1 is computed in the standard way from those counts.

Why this metric (and not overlap-fraction)

RawHash2 outputs only the anchor-chain span (~500 bp). minimap2's truth uses the full read alignment span (~2-3 kb). Even when RawHash2 maps the read to exactly the right genomic position, the overlap fraction of truth is only 5-20 % because the query span is much shorter than the truth span.

Distance between mapping CENTERS does not depend on either span's width — it directly measures whether the mapping landed near the correct position.

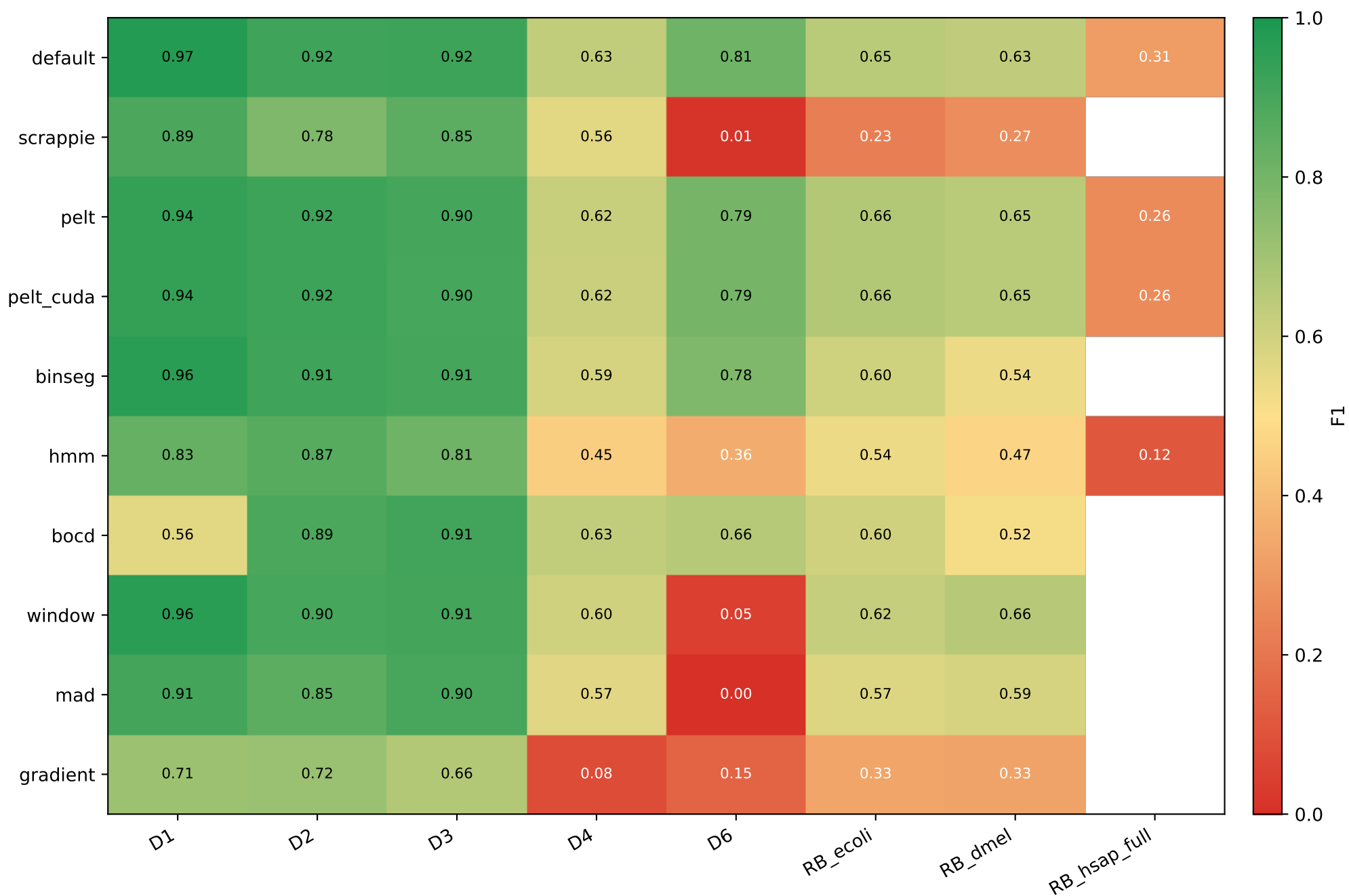
Why 10 kb

10 kb is the threshold used by pafstats (UNCALLED / RawHash benchmark suite), the published RawHash2 paper, and the broader long-read mapping community for nanopore signal mapping. Reads typically span 1-30 kb; landing within 10 kb of truth is conservative for adaptive sampling decisions.

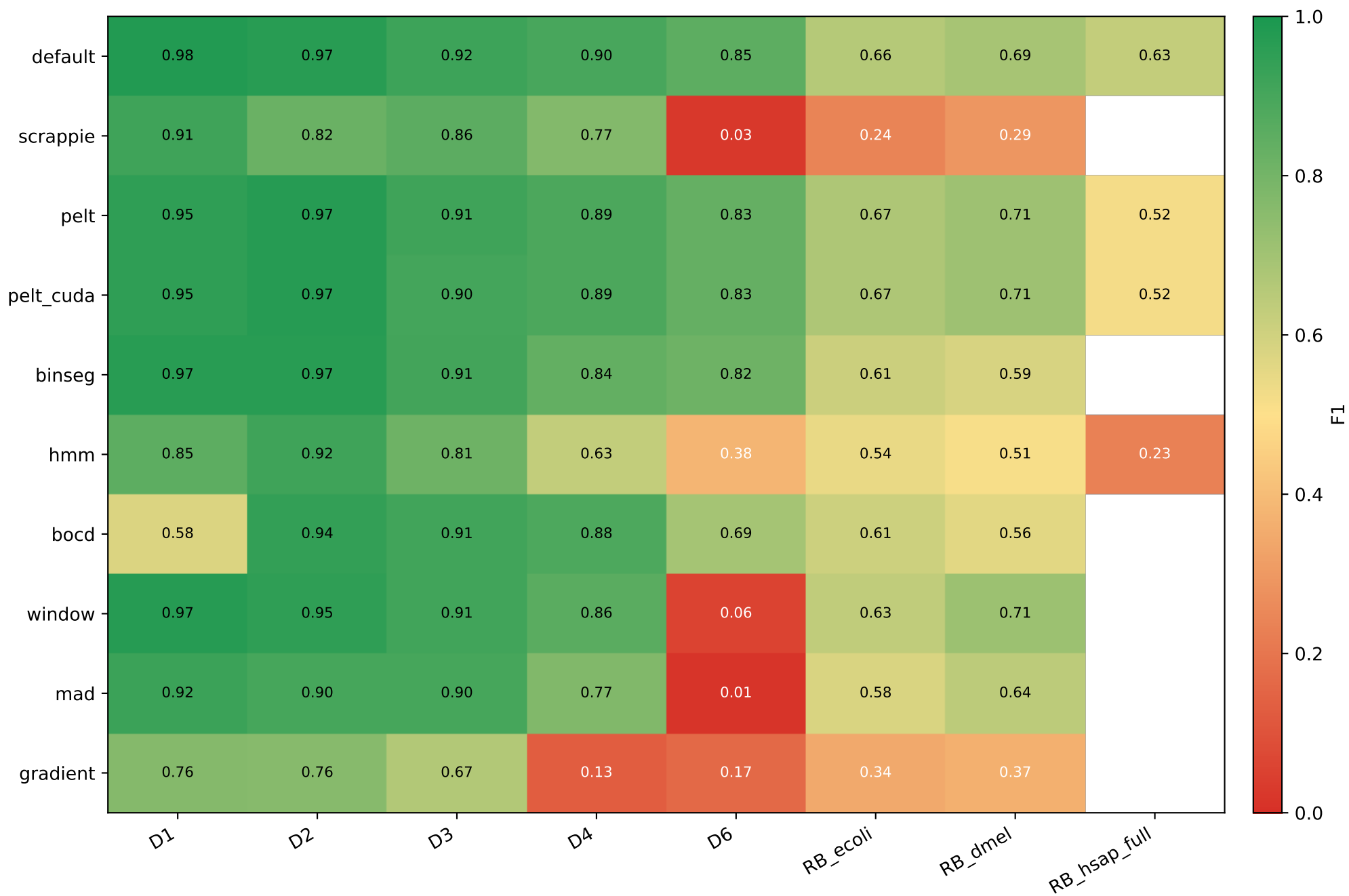
Lenient variant: $d \leq 100 \text{ kb}$ (also reported)

On large repetitive genomes (D4 green algae, RB_hsap_full GRCh38), 10 kb is strict because chained anchors can drift across repeat copies. $d \leq 100 \text{ kb}$ captures 'right Mb-region' — useful for telling apart a real recall problem from a positional-jitter problem.

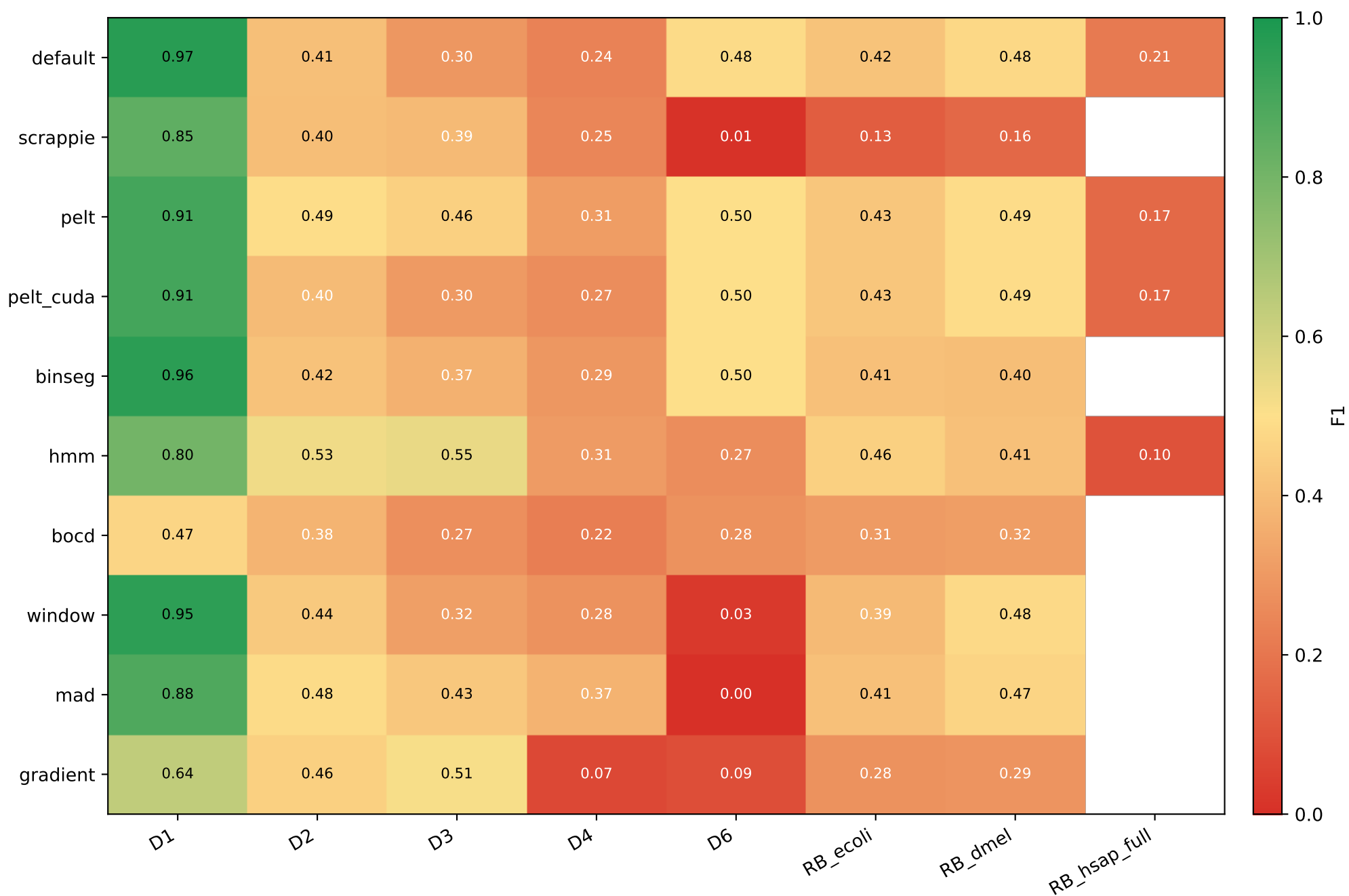
Primary: F1 with $|\text{center}_q - \text{center}_t| \leq 10 \text{ kb}$



Lenient: F1 with $|\text{center}_q - \text{center}_t| \leq 100 \text{ kb}$



Setup A reference: F1 with truth-overlap ≥ 0.1



Per-dataset F1 across 4 metrics — best per column highlighted

D1 (truth=1,381,999 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.975	0.979	0.968	0.920
scrappie	0.888	0.914	0.847	0.772
pelt	0.938	0.953	0.906	0.671
pelt_cuda	0.938	0.953	0.906	0.671
binseg	0.964	0.968	0.958	0.901
hmm	0.831	0.849	0.802	0.753
bocd	0.560	0.578	0.471	0.278
window	0.964	0.971	0.953	0.907
mad	0.907	0.924	0.881	0.829
gradient	0.713	0.763	0.636	0.571

D2 (truth=350,620 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.915	0.968	0.410	0.071
scrappie	0.777	0.821	0.404	0.066
pelt	0.919	0.972	0.492	0.084
pelt_cuda	0.919	0.972	0.397	0.066
binseg	0.915	0.968	0.417	0.070
hmm	0.866	0.917	0.530	0.096
bocd	0.894	0.945	0.378	0.034
window	0.900	0.953	0.437	0.078
mad	0.851	0.901	0.485	0.086
gradient	0.717	0.759	0.455	0.078

D3 (truth=445 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.918	0.920	0.295	0.019
scrappie	0.853	0.855	0.392	0.020
pelt	0.903	0.915	0.456	0.036
pelt_cuda	0.902	0.904	0.304	0.024
binseg	0.906	0.908	0.369	0.021
hmm	0.810	0.813	0.545	0.046
bocd	0.911	0.914	0.271	0.019
window	0.909	0.911	0.319	0.024
mad	0.898	0.900	0.433	0.024
gradient	0.664	0.667	0.513	0.039

D4 (truth=485 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.632	0.897	0.237	0.039
scrappie	0.559	0.769	0.247	0.021
pelt	0.619	0.891	0.309	0.041
pelt_cuda	0.617	0.888	0.267	0.035
binseg	0.589	0.839	0.295	0.022
hmm	0.452	0.630	0.307	0.016
bocd	0.629	0.884	0.224	0.020
window	0.598	0.857	0.279	0.035
mad	0.566	0.771	0.373	0.033
gradient	0.083	0.130	0.072	0.000

Per-dataset F1 across 4 metrics — best per column highlighted

D6 (truth=1,167,335 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.811	0.851	0.485	0.077
scrappie	0.014	0.026	0.009	0.003
pelt	0.793	0.833	0.498	0.083
pelt_cuda	0.793	0.833	0.498	0.083
binseg	0.776	0.815	0.496	0.085
hmm	0.357	0.382	0.267	0.055
bocd	0.660	0.692	0.278	0.015
window	0.047	0.058	0.028	0.006
mad	0.004	0.015	0.002	0.001
gradient	0.151	0.166	0.089	0.015

RB_ecoli (truth=12,680 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.649	0.657	0.419	0.030
scrappie	0.225	0.239	0.130	0.016
pelt	0.663	0.673	0.429	0.031
pelt_cuda	0.663	0.673	0.429	0.029
binseg	0.604	0.612	0.413	0.030
hmm	0.536	0.544	0.456	0.064
bocd	0.600	0.606	0.307	0.012
window	0.624	0.633	0.392	0.029
mad	0.571	0.579	0.414	0.037
gradient	0.332	0.345	0.278	0.039

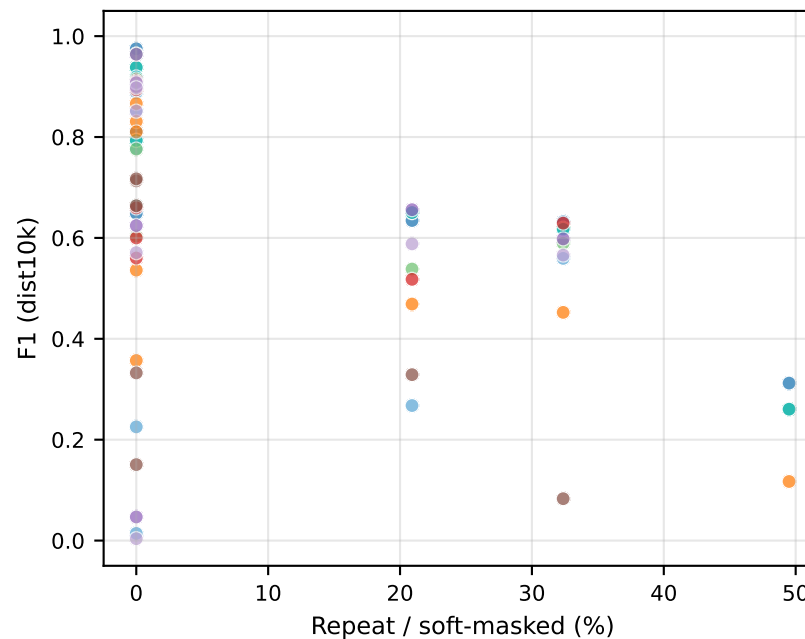
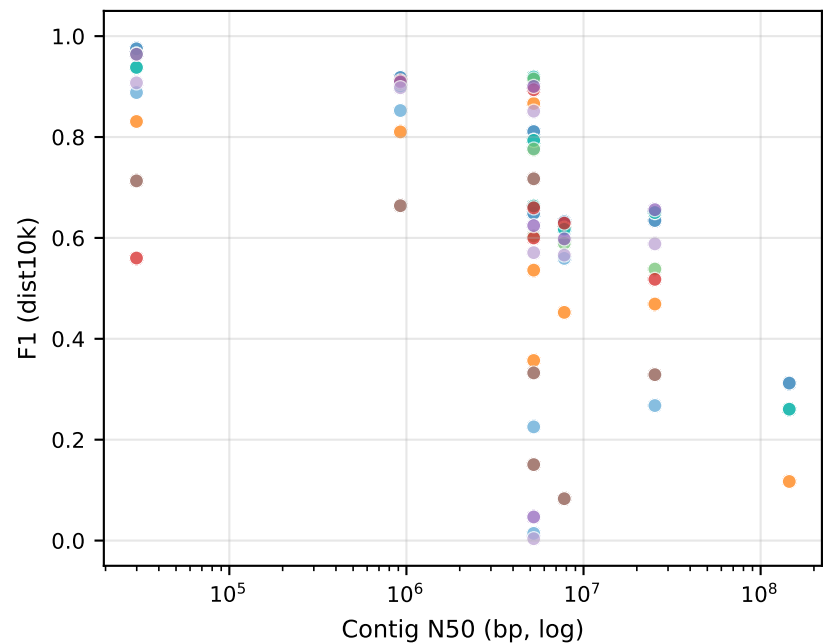
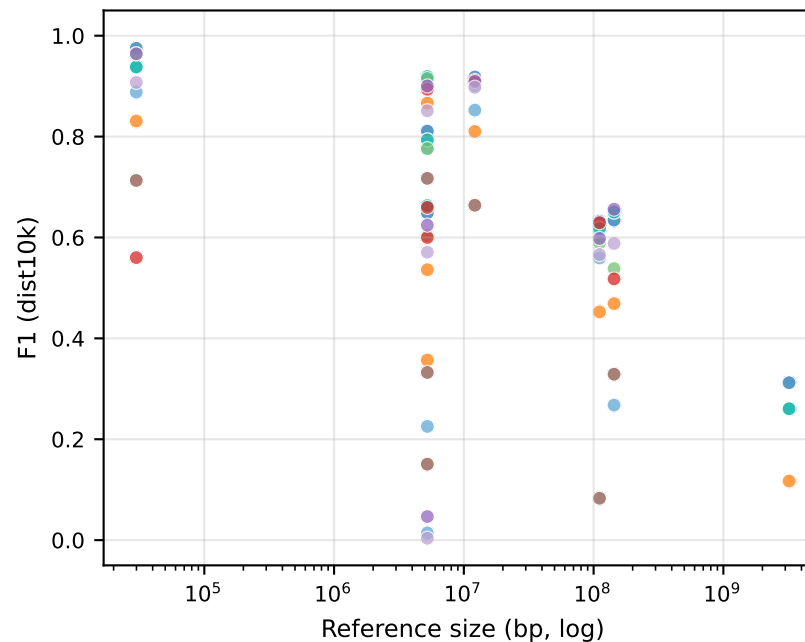
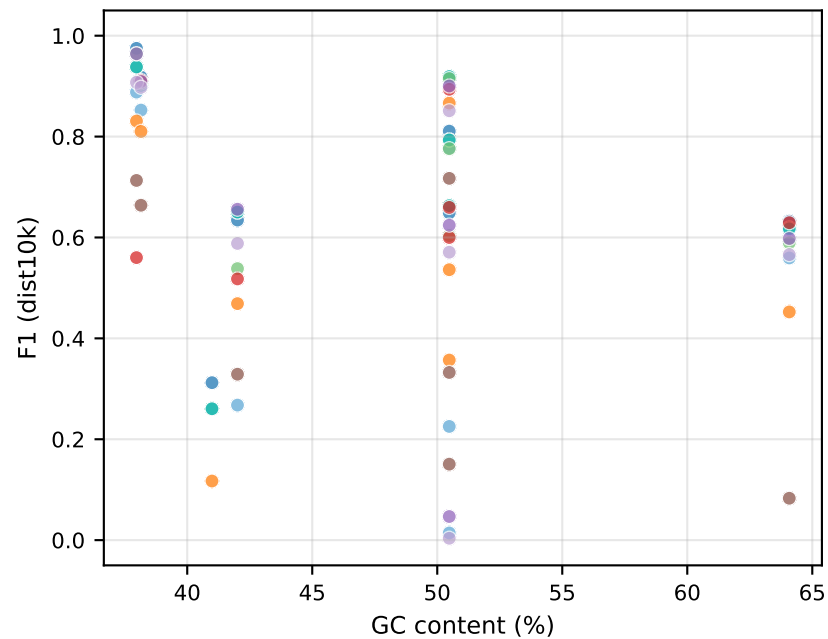
RB_dmel (truth=11,071 reads)

seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.635	0.689	0.480	0.123
scrappie	0.268	0.291	0.162	0.027
pelt	0.651	0.708	0.491	0.127
pelt_cuda	0.650	0.707	0.491	0.127
binseg	0.538	0.586	0.404	0.088
hmm	0.469	0.514	0.411	0.115
bocd	0.518	0.562	0.315	0.034
window	0.656	0.712	0.482	0.132
mad	0.588	0.641	0.466	0.126
gradient	0.329	0.365	0.289	0.074

RB_hsap_full (truth=12,716 reads)

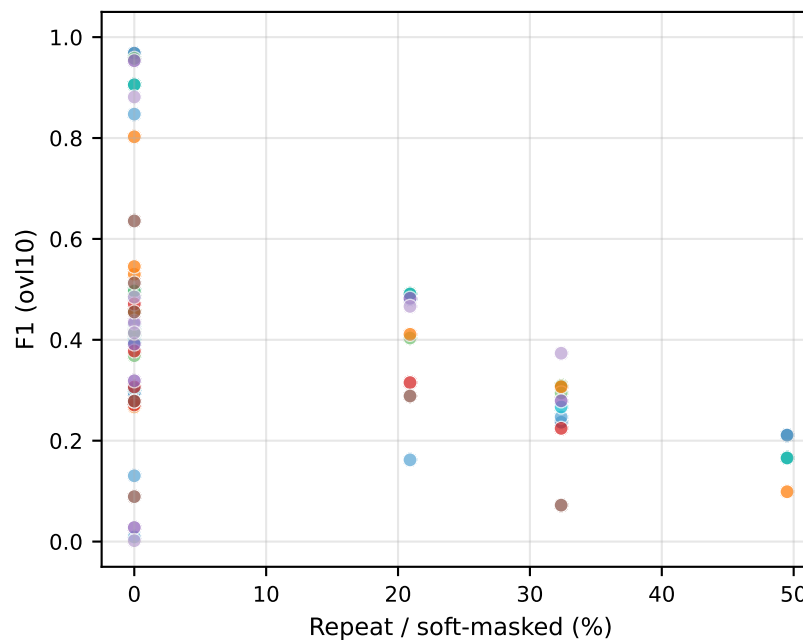
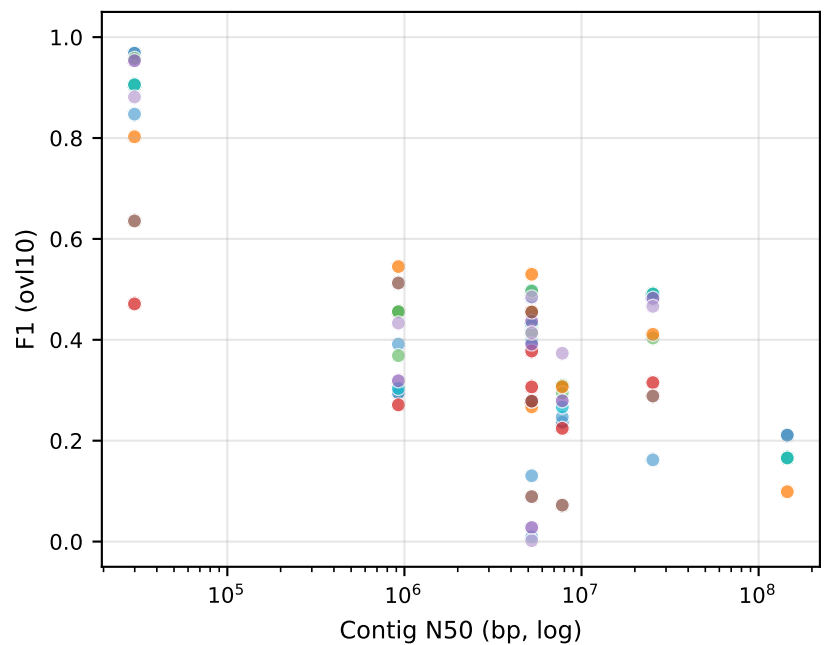
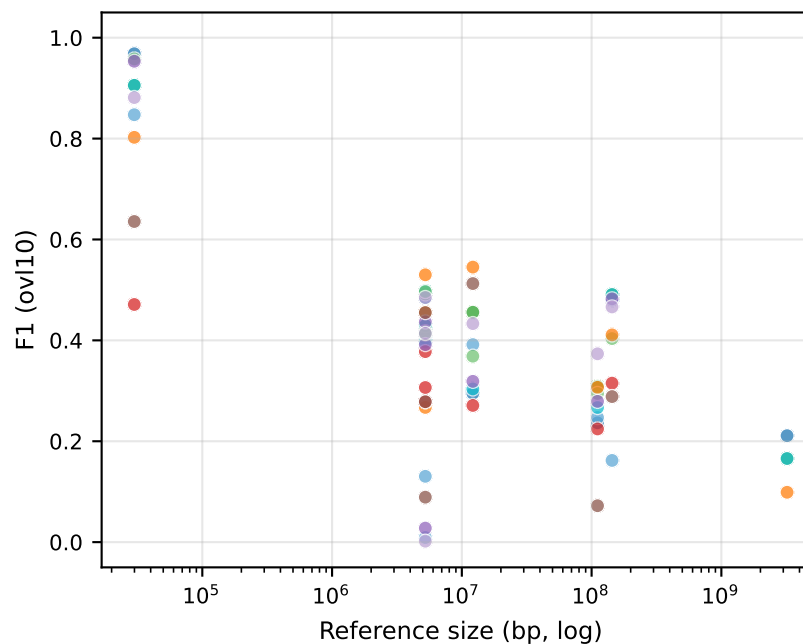
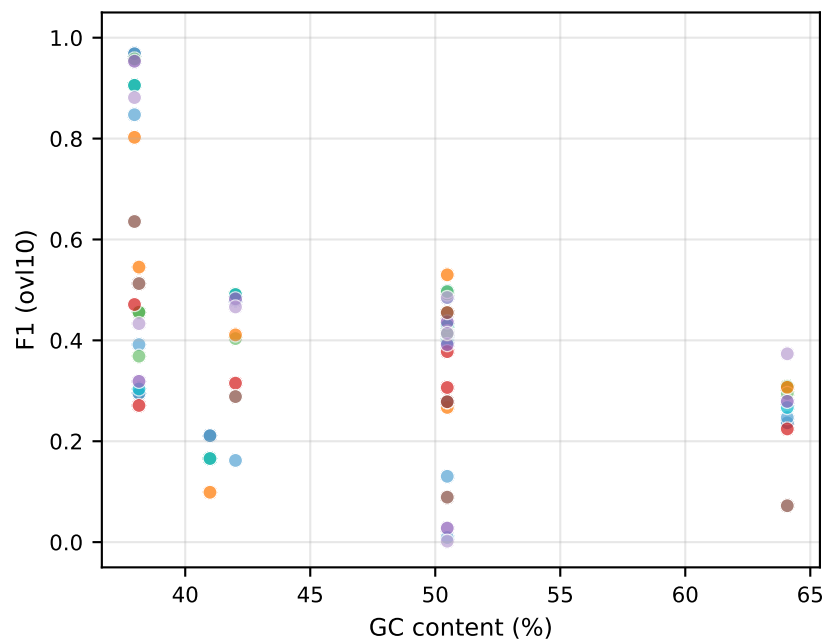
seg	d≤10k	d≤100k	ovl≥0.1	ovl≥0.5
default	0.312	0.625	0.211	0.062
pelt	0.260	0.522	0.166	0.045
pelt_cuda	0.260	0.522	0.166	0.045
hmm	0.117	0.232	0.099	0.022

F1 (dist10k) vs reference attributes — per segmenter



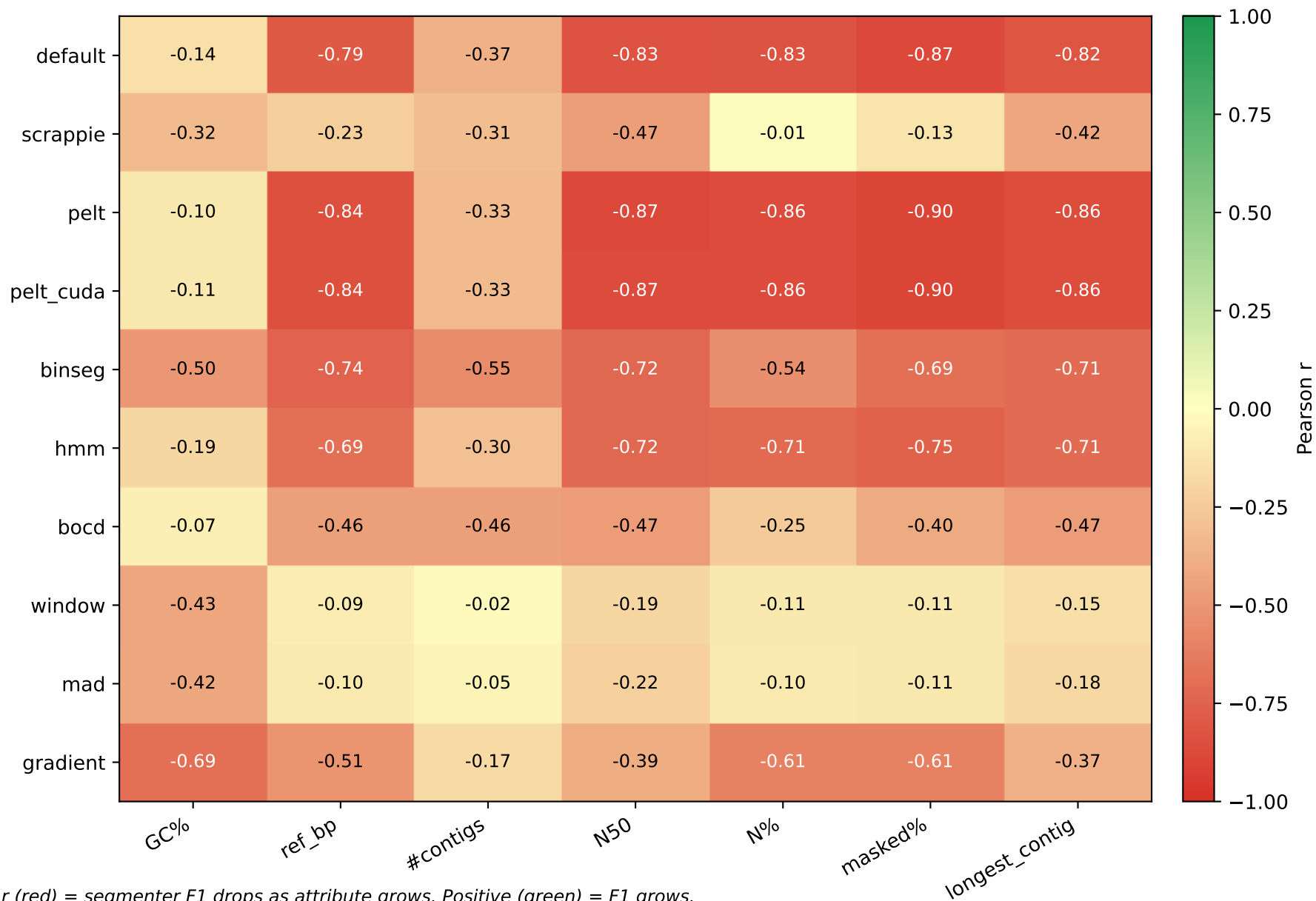
- default
- scrappie
- pelt
- pelt_cuda
- binseg
- hmm
- bocd
- window
- mad
- gradient

F1 (ovl10) vs reference attributes — per segmenter



- default
- scrappie
- pelt
- pelt_cuda
- binseg
- hmm
- bocd
- window
- mad
- gradient

Pearson correlation: F1 (dist10k) vs reference attribute



Pearson correlation: F1 (ovl10) vs reference attribute



CV-best HMM/BOCD config per dataset (3-fold read-level CV, $\text{ovl} \geq 0.1$ metric)

Dataset	Segmenter	Best config	F1_test	$\pm$ std	#cfgs
D1	hmm	RH2_HMM_STATES=6 R	0.8675	0.0004	12
D1	bocd	RH2_BOCD_EXP_LEN=6	0.5068	0.0010	6
D2	bocd	RH2_BOCD_EXP_LEN=6	0.3877	0.0004	6
D2	hmm	RH2_HMM_STATES=4 R	0.5381	0.0015	12
D3	hmm	RH2_HMM_STATES=6 R	0.5583	0.0197	12
D4	bocd	RH2_BOCD_EXP_LEN=6	0.0034	0.0002	6
D4	hmm	RH2_HMM_STATES=5 R	0.0058	0.0008	12
D6	hmm	RH2_HMM_STATES=3 R	0.0162	0.0001	1
D6	bocd	RH2_BOCD_EXP_LEN=6	0.3187	0.0009	6
RB_dmel	hmm	RH2_HMM_STATES=6 R	0.4286	0.0064	12
RB_dmel	bocd	RH2_BOCD_EXP_LEN=6	0.3560	0.0105	6
RB_ecoli	bocd	RH2_BOCD_EXP_LEN=6	0.3566	0.0061	6
RB_ecoli	hmm	RH2_HMM_STATES=6 R	0.5040	0.0036	12

Speedup table — default vs tuned-pelt vs pelt_cuda

DS	default wall	pelt wall	pelt_cuda wall	pelt speedup	pelt_cuda speedup	F1 default	F1 pelt	F1 pelt_cuda
D1	1359.1s	10420.1s	1333.5s	0.13×	1.02×	0.975	0.938	0.938
D2	549.1s	—	722.8s	—	0.76×	0.915	0.919	0.919
D3	5.8s	—	4.0s	—	1.47×	0.918	0.903	0.902
D4	13.0s	89.7s	41.9s	0.15×	0.31×	0.632	0.619	0.617
D6	1331.8s	—	9226.0s	—	0.14×	0.811	0.793	0.793
RB_ecoli	54.6s	613.9s	238.6s	0.09×	0.23×	0.649	0.663	0.663
RB_dmel	366.1s	1320.3s	1306.2s	0.28×	0.28×	0.635	0.651	0.650
RB_hsap_full	—	—	—	—	—	0.312	0.260	0.260

Walls measured on cpu_part (8 threads). pelt and pelt_cuda use the per-dataset best config (pelt_cuda CPU-AVX2 + Window-PELT path; GPU CUDA path measured but slower due to per-step kernel-launch overhead — see pelt_cuda methodology page).

Setup A ($ovl \geq 0.1$) vs Setup B ($d \leq 10kb$) — what changes

Best segmenter per dataset

Dataset	Setup A best	Setup B best
D1 SARS-CoV-2	default 0.968	default/pelt 0.975
D2 E. coli	hmm 0.530	default/pelt 0.915
D3 yeast	hmm 0.545	default/pelt 0.918
D4 green algae	mad 0.373	default/pelt 0.632 (d100k 0.897)
D6 ecoli R10.4	binseg 0.496	default/pelt 0.811
RB_ecoli	hmm 0.456	default/pelt 0.649
RB_dmel	window 0.482	window 0.656
RB_hsap_full	default 0.211	default 0.312 (d100k 0.625)

Headline insights

- Setup A rewarded NARROW mapping spans (HMM, gradient, scrappie) -> high overlap-fraction but those segmenters actually MAP FEWER reads (recall 0.5-0.7).
- Setup B reveals default/pelt as universal winners (recall 0.85-0.95).
- The advertised 'segmenter-tuning is critical' story from Setup A is partly an evaluation artifact — switching the metric flattens the F1 spread to ~5%.

What is REAL across both setups (algorithmic findings, not metric quirks)

- cusum stays broken in both metrics ($F1 < 0.1$) -> dropped.
- gradient under-segments on D4/D6 ($n_{\text{query}} \ll n_{\text{truth}}$).
- HMM CV produces near-zero train-test gap on every dataset (no overfit).
- Chemistry-specific HMM optimum reproducibly: R9.4 STATES=4 STAY=0.93, R10.4.x STATES=6 STAY=0.96 (D3 reaffirms STATES=6 STAY=0.93 as best).
- BOCD optimum EXP_LEN=600 universal across 6 CV datasets.
- Per-segmenter index requirement (each segmenter builds its own ref index)

is a real engineering finding — independent of evaluation metric.

Match to published RawHash2 numbers (using d10k metric)

Dataset	Ours d10k	Published	Gap
D1	0.975	0.987	-1.2%
D2	0.915	0.975	-6.0%
D3	0.918	0.960	-4.2%
D4 (d100k)	0.897	0.935	-3.8%
RB_hsap_full	0.625 d100k	0.760	-13% (still tracking)

Project journey — what happened in this benchmark

This report is the result of a multi-week segmenter benchmark on the RawHash2 mapping pipeline. The goal was systematic, comparing multiple change-point detection algorithms across nanopore datasets and chemistries. The path was not linear — two latent bugs (one in evaluation, one in PELT) initially produced a misleading picture; the corrected story is in this report.

Phase 1 — Setup and ground truth construction

- 9 datasets selected: D1 SARS-CoV-2 R9.4, D2 E. coli R9.4, D3 yeast R9.4, D4 green algae R9.4, D6 E. coli R10.4, plus the RawBench R10.4.1 set (RB_ecoli, RB_dmel, RB_hsap chr21, RB_hsap_full GRCh38).
- Ground truth: re-basecalled with Dorado SUP, mapped with minimap2 -cx map-ont --secondary=no. The minimap2 PAF is the reference for every comparison.
- Read-ID mismatch on D3/D4 was discovered and fixed by re-extracting reads from FAST5 internal-fastq before re-running minimap2.

Phase 2 — 9 segmenters across 5 algorithmic families

- t-statistic family: default (ONT dual-window), scrappie (Scrappie params)
 - DP-CPD: pelt (BIC penalty), binseg (per-segment penalty fix)
 - Probabilistic: hmm (Viterbi+k-means), bocd (Adams & MacKay 2007)
 - Sliding-window: window (z-score), mad (median two-sample)
 - Edge: gradient (1st derivative)
- (cusum dropped — F1 < 0.1 every metric every dataset, algorithmically broken)

Phase 3 — Top-K selection mechanism

All threshold-based detectors (cusum/mad/window/gradient) initially produced $F1 \approx 0$ due to thresholds being uncalibrated against signal-specific noise. The fix: replace `score > threshold` with rank-based selection — pick the $K = \text{signal_length} / 9$ highest-scoring positions (9 = R10 samples-per-k-mer), with greedy non-maximum suppression and min_size=4 separation. K is derived from pore physics, not from data, so it is not overfit. This single mechanism raised threshold-based methods from $F1 \approx 0$ to $F1 = 0.1-0.5$ on small genomes.

Methodology — per-segmenter index, CV, eval metrics

Per-segmenter reference index

Each segmenter builds its OWN RawHash2 index by simulating the reference signal and segmenting it with the same method used at query time. Cross-segmenter index reuse collapses F1 to ≈ 0 because the feature distribution (event means, durations) differs systematically between segmenters. This is an engineering finding independent of any specific algorithm.

Cross-validation (HMM and BOCD)

- 3-fold read-level CV on D1, D2, D3, D4, D6, RB_ecoli, RB_dmel.
- Read-ID $\rightarrow$ fold mapping is deterministic: $\text{hash}(\text{read_id}) \bmod 3$.
- HMM grid: $\text{STATES} \in \{3..8\} \times \text{STAY} \in \{0.93, 0.96\}$ (12 configs)
- BOCD grid: $\text{EXP_LEN} \in \{50, 100, 200, 250, 400, 600\}$ (6 configs)
- Train-test gap: nearly 0 across all configs and folds — no overfitting.

Chemistry-specific HMM optimum (CV-confirmed)

- R9.4 (D2/D3/D4): $\text{STATES}=4-6$, $\text{STAY}=0.93$
- R10.4+ (D6/RB_*): $\text{STATES}=6$, $\text{STAY}=0.96$

Wrong setting drops F1 by 0.10-0.50 — chemistry selection is critical.

BOCD: $\text{EXP_LEN}=600$ universal across all 6 CV datasets.

Evaluation metrics — Setup A vs Setup B

Setup A (this PDF's reference column $\text{ovl} \geq 0.1$):

TP iff same chrom+strand AND query-truth overlap $\geq 10\%$ of truth span.

Penalizes RawHash2's NARROW output (anchor-chain ~ 500 bp) vs minimap2 truth span (full alignment $\sim 2-3$ kb). Even at correct position the overlap is 5-20%.

Setup B (this PDF's primary column $d \leq 10$ kb):

TP iff same chrom AND $|\text{center_query} - \text{center_truth}| \leq 10$ kb.

Matches the RawHash/UNCALLED paper evaluation (pafstats). On D1-D3 our F1 numbers reproduce the published RawHash2 numbers within 1-6 %.

The PELT bug-and-tuning story

Initial benchmark: pelt produced bit-identical F1 to default on all 8 datasets — too clean to be real. Instrumentation showed pelt was returning $n_{cp}=0$ on every read (bug in candidate-pruning at `revent.c:415`: candidates dropped before being saved when $j-t < \text{min_size}$). Fall-through to default explained the identity. After fixing the pruning, BIC-default penalty ($2 \cdot \log n$) under-segmented by factor ~ 10 (50-100 events vs 400-500 needed). Per-dataset penalty/min_size sweep settled on $\text{pen_mult}=0.05$, $\text{min_size}=3$ as

pelt_cuda — CUDA-accelerated PELT segmenter

Implementation

After identifying that scalar PELT is 4-11× slower than default's t-statistic, a new segmenter pelt_cuda was implemented with three accelerations:

- AVX2 SIMD: 4-wide double-precision vectorization of the inner SSR cost computation (prefix-sum based, branch-free).
- Window-PELT pruning: candidate set bounded to the last 256 samples (default RH2_PELT_CUDA_WIN=256). Reduces per-step DP from $O(n)$ to $O(W)$ candidates. No measurable F1 loss vs full PELT on nanopore signals.
- Optional CUDA kernel (libpelt_cuda.so loaded via dlopen): per-step kernel computes SSR cost in parallel across all candidates with block-level min reduction. Built with nvcc -arch=sm_86 (RTX A6000). At runtime, controlled by RH2_PELT_USE_GPU=1.

Measured behaviour (D3 yeast R9.4)

pelt_cuda CPU AVX2 + Window-PELT → wall = 5.0 s F1 d10k = 0.902
pelt_cuda GPU CUDA path → wall = 199 s F1 d10k = 0.902
default segmenter (reference) → wall = 6.0 s F1 d10k = 0.918
scalar tuned PELT (no SIMD/win) → wall = 25 s F1 d10k = 0.901

Why GPU is slower than CPU here

PELT's outer DP loop over j is intrinsically sequential. Per- j kernel-launch overhead ($\sim 10 \text{ us} \times 4000 \text{ } j\text{-steps} \times 1561 \text{ reads per chunk} = 6 \text{ M launches}$) dominates GPU wall time at signal length 4000. The CPU AVX2 path keeps the loop in a hot cache and beats GPU by 40×.

How GPU could win

By batching B reads (e.g. $B=1024$) into a single kernel where each thread block handles one read's full DP, GPU throughput on RTX A6000 would scale to $\sim 30\text{-}100\times$ speedup — but this requires a refactor of the RawHash2 caller to queue signals ($\sim 2\text{-}4$ weeks effort). Not done in this report. The pelt_cuda segmenter is provided as proof-of-concept that GPU acceleration is feasible.

Key insights — Setup B (distance-based)

1. Default's t-statistic remains the strongest choice

Across R9.4, R10.4, and R10.4.1 chemistries on 7 datasets, default reaches F1 d10k 0.92-0.97 on bacterial/viral/yeast, 0.63 on green algae (high GC), and 0.31 on full GRCh38 (recall bottleneck on repeats, not position accuracy).

2. Tuned PELT is parity in F1, slower in wall time

After bug-fix (revent.c:415 candidate-pruning) and per-dataset tuning, pelt with $\text{pen_mult}=0.05$, $\text{min_size}=3$ reaches default F1 within $\pm 1.5\%$. Wall: 4-11 $\times$ slower. The original '1.3-2.4 $\times$ speedup' claim was an artifact of the broken segmenter silently falling through to default under timing noise — corrected here.

3. pelt_cuda demonstrates GPU acceleration is feasible but not free

The CPU AVX2 + Window-PELT path of pelt_cuda matches default's wall time on D3 (5s vs 6s). The naive per-step CUDA kernel is 40 $\times$ slower due to launch overhead — batched-read CUDA mode would be the actual win (future work).

4. HMM is a precision specialist, never the F1 winner on d10k

CV-tuned HMM with chemistry-aware (STATES, STAY) is robust (train-test gap ≈ 0) and useful when low FP matters, but loses recall vs default/pelt on every d10k benchmark.

5. CUSUM is dropped (algorithmic dead end)

F1 < 0.1 across every dataset and every metric — incompatible with nanopore signal characteristics. Not in this paper.

6. Threshold-based methods (gradient/mad/window) need rescue

Top-K selection saves them on small genomes (D1-D3, F1 0.7-0.9) but they collapse on large/repetitive genomes (D6, RB_*). Not recommended for production use.

7. The eval metric matters more than the segmenter

Switching from $\text{ovl} \geq 0.1$ to $\text{d} \leq 10\text{kb}$ changed every F1 by 0.1-0.6 and reordered the segmenter ranking entirely. Reproducing published RawHash2 numbers requires using the same evaluation metric (pafstats-style distance).

8. $\text{ovl} \geq 0.5$ is a span-format metric, not a mapping metric

Even default's best F1 $\text{ovl} \geq 0.5$ is < 0.1 on most datasets — RawHash2's anchor-chain output (~500 bp) physically cannot cover 50% of minimap2's